STADIOEquities
Part C – Data Request

Data Science Capstone Project

Project focus: Predicting customer activation and early dormancy. The requested data will support developing a data-driven model to identify customers likely to activate their accounts or become dormant.

Table 1: Customer Registration and Demographic Data

Column name	Data type	Example	Additional information
Customer ID	String	CUST001245	Unique anonymised customer identifier. Must remain consistent across all datasets.
Registration date	Datetime	2026-03-15 10:25:00	Date and time the customer registered.
Age	Integer	28	Customer age at registration.
Region	Categorical	Gauteng	Customer province/region. Use a consistent list of valid regions.
Stated goal	Categorical	Long-term investment	Investment goal selected by the customer during registration.
Risk appetite	Categorical	Moderate	Use consistent categories such as Low, Moderate and High.
Acquisition channel	Categorical	Social media	Channel through which the customer was acquired.
Campaign	String/Categorical	Campaign_2026_01	Marketing campaign associated with acquisition.
Acquisition cost	Numeric	180.00	Cost incurred to acquire the account, where available.
Device type	Categorical	Mobile	Valid examples: Mobile, Tablet, Desktop.

Table 2: Account, KYC and Onboarding Data

Column name	Data type	Example	Additional information
Account ID	String	ACC008921	Unique account identifier linked to Customer ID.
KYC start date	Datetime	2026-03-15 10:30:00	Date and time KYC started.
KYC completion date	Datetime	2026-03-15 11:05:00	Date and time KYC was completed. Blank if not completed.
KYC status	Categorical	Completed	Valid values should include Started, Completed, Failed and Abandoned.
Onboarding steps	Integer	8	Number of onboarding steps completed.
Onboarding abandonment	Binary	1	1 = abandoned; 0 = completed.

Column name	Data type	Example	Additional information
Abandonment step	Categorical	Identity verification	Specific onboarding step where the customer stopped.
Registration status	Categorical	Active	Current account status, using the client's standard status definitions.
Days to first deposit	Integer	5	Number of days between registration and first deposit. Blank if no deposit occurred.

Table 3: App and Web Behaviour

Column name	Data type	Example	Additional information
Session ID	String	SES102938	Unique identifier for each app/web session.
Customer ID	String	CUST001245	Must link to the customer dataset.
Session date	Datetime	2026-03-16 14:32:10	Date and time of the session.
Platform	Categorical	Android	Examples: Android, iOS, Web.
Session duration	Numeric	12.5	Session duration in minutes.
Screens viewed	Integer	14	Number of screens viewed during the session.
Features used	Integer	4	Number of platform features used during the session.
Login frequency	Integer	8	Number of logins during the defined observation period.
Onboarding interaction	Binary	1	1 = interacted with onboarding; 0 = no interaction.
Deposit page viewed	Binary	1	1 = viewed deposit functionality; 0 = not viewed.

Table 4: Account Funding and Transaction Data

Column name	Data type	Example	Additional information
Transaction ID	String	TXN0005678	Unique transaction identifier.
Customer ID	String	CUST001245	Links the transaction to the customer.
Transaction date	Datetime	2026-03-20 09:15:00	Date and time of transaction.
Transaction type	Categorical	Deposit	Examples: Deposit, Withdrawal.
Transaction amount	Numeric	1500.00	Monetary value of transaction. Currency should be specified in the data dictionary.
Deposit method	Categorical	EFT	Method used to fund the account.
Account balance	Numeric	3500.00	Balance after transaction, where available.
First deposit	Binary	1	1 = transaction is the customer's first deposit; 0 = otherwise.
Withdrawal amount	Numeric	500.00	Amount withdrawn, where applicable.

Table 5: Trading Activity

Column name	Data type	Example	Additional information
Trade ID	String	TRD004562	Unique trade identifier.
Customer ID	String	CUST001245	Links the trade to the customer.
Trade date	Datetime	2026-04-02 13:45:00	Date and time of trade.
Trade type	Categorical	Buy	Valid values: Buy, Sell.
Instrument type	Categorical	ETF	Type of investment instrument.
Trade value	Numeric	2500.00	Monetary value of trade.
Number of trades	Integer	12	Number of trades within the defined observation period.
Portfolio concentration	Numeric	0.72	Proportion of portfolio concentrated in the largest holding; preferably 0–1.
Investment frequency	Numeric	3	Average number of investment transactions per defined period.

Table 6: Marketing and Customer Engagement

Column name	Data type	Example	Additional information
Customer ID	String	CUST001245	Unique anonymised customer identifier.
Campaign ID	String	CAMP2026A	Marketing campaign identifier.
Message date	Datetime	2026-03-18 08:00:00	Date and time the message was sent.
Message type	Categorical	Email	Examples: Email, SMS, Push notification.
Message opened	Binary	1	1 = opened; 0 = not opened.
Link clicked	Binary	1	1 = clicked; 0 = not clicked.
Nudge response	Binary	0	1 = responded/interacted; 0 = no response.
Number of nudges	Integer	3	Number of engagement interventions received during the observation period.

Table 7: Customer Support and Feedback

Column name	Data type	Example	Additional information
Ticket ID	String	TKT005678	Unique support ticket identifier.
Customer ID	String	CUST001245	Links ticket to customer.
Ticket date	Datetime	2026-04-12 15:20:00	Date and time ticket was submitted.
Ticket reason	Categorical	Deposit issue	Standardised reason for customer contact.
Resolution time	Numeric	24	Time taken to resolve ticket, preferably in hours.

Column name	Data type	Example	Additional information
Customer sentiment	Categorical	Negative	If sentiment analysis is already available; otherwise provide raw text separately where permitted.
Complaint indicator	Binary	1	1 = complaint; 0 = normal enquiry.

Table 8: Target Variable – Activation and Dormancy

Column name	Data type	Example	Additional information
Customer ID	String	CUST001245	Unique anonymised customer identifier.
First_Deposit_30_Days	Binary	1	1 = first deposit within 30 days; 0 = no first deposit within 30 days
First deposit date	Datetime	2026-03-20	Date of customer's first deposit.
Active 90 days	Binary	1	1 = customer traded or held investments within the last 90 days; 0 = otherwise.
Dormant 6 months	Binary	0	Primary target: 1 = account became dormant within six months; 0 = remained active.
Dormancy date	Datetime	2026-09-20	Date the account became dormant, where applicable.
Reactivation indicator	Binary	0	1 = dormant customer subsequently became active; 0 = not reactivated.

Project Target

The primary target variable is customers who make their first deposit within 30 days and become dormant within 6 months. The project should investigate whether customer demographic, onboarding, behavioral, funding, trading, and engagement data can predict which customers are likely to become dormant within six months.

Data Quality and Delivery Requirements

- Provide historical records covering both customers who made their first deposit within 30 days and customers who became dormant within 6 months.
- Use a consistent anonymised Customer ID across all datasets so that sources can be joined reliably.
- Provide dates and timestamps in a consistent format, preferably YYYY-MM-DD HH:MM:SS.
- Clearly identify missing, unknown, and not-applicable values; do not replace missing values with arbitrary numbers.
- Provide a data dictionary for all fields, including definitions, valid categories, units, and currency.
- Identify duplicate records and explain any known data-quality issues before delivery.

- Provide sufficient historical data to support training, validation, and testing of predictive models.
- Where free-text or compliance data are supplied, ensure that privacy, confidentiality, and applicable data-protection requirements are addressed.

Project Question

“Can STADIOEquities use customer demographic, onboarding, behavioral, funding, trading, and engagement data to accurately predict which accounts are most likely to make their first deposit within 30 days and which active accounts are at risk of becoming dormant within six months?”